

Skills Summary

Interpreting the Solution of a System on a Graph

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the solution off a graph

What a solution is: an ordered pair (x, y) that makes *both* equations true. On a graph that is the point where the two lines cross — the only point lying on both. A point on just one line solves that equation alone, not the system. Always write the answer as a pair (x, y) and check it in *both* equations.

Example: Read the solution of $y = x + 2$ and $y = -2x + 8$ from their graph.

Step 1. A solution has to sit on both lines, so I look for the single crossing point rather than solving anything.

Step 2. The lines cross at $x = 2$; substituting gives $y = 2 + 2 = 4$ on the first and $y = -4 + 8 = 4$ on the second, so both agree.

$\Rightarrow (2, 4)$

Try it: Where do $y = x + 1$ and $y = -x + 7$ cross?

check: $(3, 4)$

Skill 2 — One solution, none, or infinitely many

Read it off the slopes and intercepts:

- Different slopes: the lines cross once — *exactly one solution*.
- Same slope, different y -intercepts: parallel, never meeting — *no solution*.
- Same slope *and* same intercept: one line drawn twice — *infinitely many*.

Put both equations in $y = mx + b$ form before comparing.

Example: How many solutions has $y = 3x + 2$ and $y = 3x - 4$?

Step 1. Both are already in slope-intercept form, so I compare slopes first — that is what decides the number of crossings.

Step 2. The slopes are both 3, so the lines are parallel; the intercepts 2 and -4 differ, so they are not the same line.

Step 3. Parallel and distinct lines never meet.

$\Rightarrow$ no solution

Try it: How many solutions has $y = -x + 5$ and $y = -x + 1$?

no solution check:

Skill 3 — Graphing a system to solve it

How: rewrite each equation as $y = mx + b$, plot the y -intercept, count the slope as rise over run to a second point, and draw the line. Do that for both, then read the crossing point. A graphical reading is exact only when the crossing lands on a lattice point — otherwise use it as an estimate and confirm by substitution.

Example: Solve $y = -x + 6$ and $2y = x$ by graphing.

Step 1. Neither equation is in a form I can plot directly from a second point, so I put both into slope-intercept form first.

Step 2. The first is already $y = -x + 6$: intercept 6, slope -1 . The second becomes $y = \frac{1}{2}x$: through the origin, slope $\frac{1}{2}$.

Step 3. The lines meet where $-x + 6 = \frac{1}{2}x$, so $x = 4$ and $y = 2$ — a lattice point, so the graph can be read exactly.

$\Rightarrow$ **(4, 2)**

Try it: Solve $y = -x + 5$ and $y = 2x - 1$ by graphing.

(8, 3) check: